

The Ambiguous Brief

Builds: Working with Ambiguity · Decomposition

A turnkey unit for teaching students to ask first and plan before they build.

Builds: Working with Ambiguity · Decomposition **Grade band:** 6–12 (adaptation notes for grades 3–5 and college below) **Time:** 3 sessions of ~45 minutes, run as three iterations across a term (not three days in a row) **Format:** Whole-class launch → small-group work → individual cold submission by the third iteration

Why this unit

This is the silence from the keynote. When a real project lands in the workforce, nobody hands you a rubric — they hand you a thin brief and expect you to ask the questions that turn it into a plan. Our students have been trained to wait quietly and panic the night before. This unit makes asking questions *early* the assignment itself, and builds the muscle of turning an open prompt into ordered, executable steps. It is the single highest-leverage unit in the set, because Working with Ambiguity and Decomposition sit underneath everything else.

Learning objective

By the end of the third iteration, **students can ask targeted clarifying questions before beginning an open-ended project, and independently break the prompt into a concrete, ordered, time-paced plan** — without a teacher-provided checklist.

Materials & prep

Teacher prep (~20 min the first time, ~5 min after):

- Write one deliberately *thin* brief per iteration. A good brief states a goal and an audience but withholds the format, the constraints, and the success criteria on purpose. Example: *"Create something that helps a newcomer understand this community."* Keep a running bank of 4–6 briefs so you can rotate them.
- Print the **Student Handout** (below), one per student, or post it to your LMS.
- Be ready to play "the client" — the person who commissioned the work and is the only one who can answer questions. Decide in advance what your answers will be so you stay consistent across groups.

Students need: the handout, something to write with (pencil and paper is the point — see the keynote), and for iteration 3, the question checklist they helped build in iteration 2.

A brief bank you can use

Each of these is deliberately *thin* on purpose — it names a goal and an audience but withholds the format, the length, the materials, and the definition of "done." That missing information is exactly what students must surface with questions. Hand one out as-is (don't add detail), and adapt the bracketed parts to your subject. Rotate a different brief through each iteration so students can't reuse last time's plan.

1. **The newcomer.** *"Create something that helps a new student feel at home in our school."*
2. **Teach it down.** *"Make something that teaches a younger student one important idea from [our current unit]."*
3. **Make the case.** *"Propose a change that would make our classroom or school work better — and convince the person who could actually approve it."*

4. **The local issue.** *"Help people in our town understand [a topic we've studied] and why it matters to them."*
5. **The how-to.** *"Create a resource that would let someone complete [a task from this class] without your help."*
6. **Get them to care.** *"Get a specific group of people to care about [a topic from this course]."*

The one rule when you hand it over: if a student asks "what format should it be / how long / what counts as good?", don't answer it as fact — answer as the client ("That's a great question — what do you think serves the audience best?") or hand the decision back. The point is that *they* resolve the ambiguity, not that you fill it in.

The arc at a glance

Session	Focus	~Time
1 — Surface	Read the thin brief; write down assumptions and every question needed before starting	45 min
2 — Interrogate	Structured client Q&A; turn answers into an ordered, paced plan	45 min
3 — Cold start	Submit a clarifying-questions doc + plan with no scaffold; self-assess	45 min

Step-by-step sequence

Session 1 — Surface the unknowns (45 min)

- **(5 min) Frame the rules.** Tell students plainly: *"This brief is incomplete on purpose. Your first job is not to start building — it's to figure out what you don't know yet. Part of your grade is the questions you ask."* This permission is the whole move.
- **(10 min) First-instinct capture.** Hand out the brief. Students write, silently, the first thing they'd do if they had to start *right now*. They keep this — it's the "before" half of the discussion artifact.
- **(20 min) Assumption + question dump.** Individually, then in pairs, students list (a) every assumption they're making about the brief, and (b) every question they'd need answered before they could confidently begin. Push for quantity. Iteration 1 only: model 3–4 strong clarifying questions on the board to prime the pump ("Who exactly is the audience?" "What does 'understand' look like — a tour, a guide, a video?" "How will we know it worked?").
- **(10 min) Sort.** Groups sort their questions into *must-know-before-I-start* vs. *can-figure-out-as-I-go*. Collect the must-know lists; you'll answer them as the client next session.

Session 2 — Interrogate, then plan (45 min)

- **(15 min) Client Q&A.** You are the client. Groups ask their must-know questions; you answer consistently. Refuse to answer questions they could reasonably decide themselves — say *"That's your call to make and defend."* This is where they learn the difference between a clarifying question and offloading a decision.
- **(20 min) Decompose into a plan.** Groups turn the now-clarified brief into an ordered plan: phases, a rough owner per phase, and a rough time estimate per phase. Require them to name **the blocker** — the one step that must happen first. (This is the Decomposition rubric, row 2.)
- **(10 min) Before/after debrief.** Each group reads their Session-1 "first instinct" next to their Session-2 plan. The gap *is* the lesson. Ask: "What did asking questions change about where you would have started?"

Session 3 — Cold start (45 min) — iteration 3 only, fully released

- **(5 min) No scaffolding.** Hand out a new thin brief. No modeling, no pairs, no priming.

- **(25 min) Individual submission.** Each student submits, on their own: a clarifying-questions document (sorted must-know vs. later) and an ordered, time-paced plan naming the blocker.
- **(15 min) Self-assessment.** Students score themselves against the checklist they co-wrote in iteration 2 and against the rubric language. Collect both the work and the self-assessment.

Student handout (copy-ready)

The Ambiguous Brief

You've just been handed a real-world brief. It is *supposed* to feel incomplete — that's how real projects arrive. Your job today is **not** to start building. Your job is to figure out what you don't know, and to ask for it.

The brief: *(your teacher will give you this)*

Step 1 — First instinct. In two sentences: if you had to start *right now*, what would you do first? (Keep this. We'll come back to it.)

Step 2 — Assumptions. List every assumption you're making about this brief. ("I'm assuming the audience is...", "I'm assuming this should be a...")

Step 3 — Questions. List every question you'd need answered before you could confidently begin. Aim for at least eight. Then mark each one:

- **(M)** must know before I start
- **(L)** can figure out later

Step 4 — The plan. After the client answers your (M) questions, write your plan:

1. The phases, in order
2. The one step that must happen *first* (the blocker) — and why
3. A rough time estimate for each phase

A good clarifying question removes a real unknown ("Who is the audience?"). **A weak one** asks someone else to make your decision for you ("What color should it be?"). Aim for the first kind.

What proficient looks like (teacher look-fors)

- The student asks **targeted** questions in the first session — not "what do you want?" but "who is this for, and what does success look like?"
- Questions arrive **at the start**, not the night before.
- The plan is **ordered**, names a **blocker**, and **paces** the phases against available time.
- When you change something as the client, the student **adjusts the plan** instead of restarting.

Assessment

Score with [toolkit/rubrics.md](#), two clusters, no separate grade required — lay them over the work students already produced:

- **Working with Ambiguity** → *"...ask clarifying questions early, not the night before something is due." and "...make a stated assumption and keep working when information is incomplete."*
- **Decomposition** → *"...break a project into ordered steps without being handed a checklist." and "...identify which step must come first (the blocker)."*

Proficient = does it independently this time. **Advanced** = anticipates the ambiguity before you surface it, or drafts a working definition to validate early.

The rubric for this unit

Lay these competency rows over the work students already produced — no separate assignment. **Proficient** is the target (demonstrated and independent); **Advanced** means the student extends the skill unprompted.

Working with Ambiguity

The student can...	Not Yet	Approaching	Proficient	Advanced
...ask clarifying questions early, not the night before something is due.	Goes silent when a project is introduced; sends questions only when the deadline is hours away.	Asks some questions but waits until stuck rather than front-loading key clarifications at the start.	Within the first day or two, asks targeted questions that remove the biggest unknowns before starting.	Anticipates ambiguities before the teacher surfaces them and drafts a working definition to validate early.
...make a stated assumption and keep working when information is incomplete.	Freezes or refuses to proceed without explicit instructions for every step.	Will move forward when told it is okay, but does not name or flag the assumption made.	States the assumption out loud ("I'm treating X as Y — let me know if that's wrong") and keeps working.	Names assumptions, weighs the risk of each, and designs the work so it can adapt quickly if an assumption is wrong.
...adapt when requirements shift mid-project.	Treats a change in scope as a crisis; restarts from scratch or shuts down.	Adjusts the work but needs reassurance before continuing and may discard still-usable material.	Acknowledges the change, identifies which parts of existing work still apply, and updates the plan without drama.	Frames the change as useful data, identifies the delta, adjusts with minimal lost work, and brings the team along.

Decomposition

The student can...	Not Yet	Approaching	Proficient	Advanced
...break a project into ordered steps without being handed a checklist.	Waits for the teacher to provide a step-by-step list; cannot name a next action independently.	Names a few steps when prompted, but order is incomplete or missing key pieces.	Independently maps out an ordered sequence before asking for help, and adjusts the plan when something changes.	Produces a sequenced breakdown unprompted, flags dependencies or risks, and helps a peer do the same.
...identify which step must come first (the blocker).	Lists tasks randomly with no sense of what must happen before something else can start.	Puts most steps in a workable order but skips at least one dependency (e.g., writing before researching).	Sequences steps so each one sets up the next and can explain why a particular step is the blocker.	Identifies the critical path and redesigns the sequence when a blocker changes.
...size each step so work can be paced.	Treats all tasks as the same effort; no sense of relative size.	Can identify big vs. small tasks but does not connect estimates to available time.	Sizes each step well enough to decide where to start and how to pace across available days.	Estimates effort, builds in buffer for likely problems, and revises estimates mid-project based on evidence.

Gradual release across the three iterations

	Teacher provides	Student provides
Iteration 1	Models strong questions; pairs allowed; client answers generously	Questions with heavy support

	Teacher provides	Student provides
Iteration 2	No modeling; pairs review before the client meeting; checklist co-written	Questions independently, plan in groups
Iteration 3	Nothing — a cold brief	Questions + plan submitted solo; self-assessed against the co-written checklist

The goal is that by iteration 3, asking the brave question early is *automatic*, not assigned.

Differentiation & adaptation

- **Grades 3–5:** Use a concrete, familiar brief ("plan a class party") and keep the question list to four. Skip time estimates; keep the must-know/later sort.
- **College / CTE:** Use a real client brief from industry or a capstone, and require a written assumptions log they revise as the project proceeds.
- **Multilingual learners / IEPs:** Provide question *stems* ("Who is...?", "How will we know...?", "What if...?") rather than a list of questions — the stems scaffold the form without giving away the content.
- **Time-crunched version:** Run only Session 1 and Session 2 as a single 50-minute block on the launch day of a project you were already assigning. That alone breaks the silence.

Common pitfalls

- **Answering decisions, not just unknowns.** When you over-answer as the client, you rebuild the checklist. Hand decisions back.
- **Grading the questions for being "right."** Grade them for being *early*, *targeted*, and *sorted* — not for matching an answer key.
- **Running the three sessions back-to-back.** The skill builds by *spacing* the iterations across a term so the scaffolding can actually withdraw.

The low-lift version (if you have no room for the full unit)

Take any assignment you already give, delete the most specific instruction from it, and tell students up front: *part of your grade is the clarifying questions you ask in the first ten minutes*. That's the [Grade the Questions](#) weave-in move — the entire unit compressed into one sentence of permission.